Tristan Sand

tristan.sand@northwestern.edu | 0009-0004-5934-9650 | tristan-sand.github.io

Education

Northwestern University Sep. 2026 – Present
PhD in Astronomy *Evanston, IL*

Loyola University Chicago Aug. 2024 – May 2026
Bachelor of Science in Physics with Computer Science and Mathematics – Summa Cum Laude *Chicago, IL*

Drake University Aug. 2022 – May 2024
Bachelor of Science in Physics, Astronomy, and Mathematics – Dean's List *Des Moines, IA*

Research Experience

Center for Interdiciplinary Exploration and Research in Astrophysics REU June 2025 – Aug. 2025
Northwestern University *Evanston, IL*

- Conducted research focused on stellar collisions, dense galactic clusters, and machine learning.
- Participated in multiple career development programs, such as RCTP (research communication training program) and Python workshops.
- Prepared a writing deliverable, website, and poster related to project results.
- Completed research project over 2025-2026 school year.

MISSFIT Student Research Lead Jan. 2023 – July 2024
Drake University *Des Moines, IA*

- Developed the artificial gravity project for the student-led MISSFIT (magneto-ionization spacecraft shielding for interplanetary travel) collaboration.
- Designed 3D printed proof of concept for a new approach to artificial gravity generator.
- Organized team schedule and meetings and designated projects to other members.
- Participated in summer research program during 2023 and 2024 Summer.
- Assisted in work of other subgroups, such as the computational and biological subgroups.

Publications and Presentations

Sand T.C, Rose S.C., González Prieto E., Lombardi J.C., Gibson C.F.A., Kıroğlu F., Kremer K., O'Conner C.E., Ramirez-Ruiz E., Rasio F.A., and Starkenburg T. "Using Machine Learning to Model Stellar Collisions in the Galactic Center", 2026, Preprint, arXiv:2609.25318, 247th AAS Poster Presentation

González Prieto E., Lombardi J.C., Rose S.C., Gibson C.F.A., O'Conner C.E., Starkenburg T., Kıroğlu F., Kremer K., **Sand T.C**, and Rasio F.A., "Machine Learning Methods for Stellar Collisions. I. Predicting Outcomes of SPH Simulations", 2026, ApJ, 1005, 131

S. Chen, M. Holcomb, E. Jackovin, X. Kwa, J. Kilburg, A. Lobo, J. Messerli-Wallace, E. Miller, S. Mortenson, **T. Parmerlee**, G. Summers, M. Tentis, K. Weber, M. Viscarra; T. Kutnink (Mentor), M. Lutrell (Mentor), A. Petridis (Mentor), "MISSFIT: Artificial Gravity Generation through High-Frequency Oscillations", 2024, APS Poster Presentation & ISGC Undergraduate Research Symposium Presentation

Awards, Fellowships, and Honors Societies

Math Department Chair's Award Loyola University Chicago, 2026
Physics Department Service Award Loyola University Chicago, 2026
Alpha Sigma Nu Honor Society Induction Loyola University Chicago, 2025
Rataj Mathematics Scholarship Loyola University Chicago, 2025-2026
Loyola University Chicago Mulcahy Fellowship (Declined) Loyola University Chicago, 2025-2026
Sigma Pi Sigma (Physics Honor Society) Loyola University Chicago, 2025
Father Gerst Memorial Award Loyola University Chicago, 2024-2025
Putnam Math Competition Rank of 2566 out of 3857 MAA Putnam Competition, 2023
ISGC Undergraduate STEM Research Scholarship Iowa Space Grant Consortium, 2023-2024
Drake Physics Prize Runner-Up Scholarship Drake University, 2022-2024

Relavent Course Work

Northwestern University Sep. 2026 – Present

- Classical Mechanics, Astrophysical Radiative Processes and Transport, Machine Learning: Foundations, Applications, and Algorithms

Loyola University Chicago Aug. 2024 – May 2026

- Freshman Projects, Electronics, Optics, Theoretical Mechanics, Electricity and Magnetism, Thermal Physics and Statistical Mechanics, Quantum Mechanics, General Relativity and Cosmology, Undergraduate Research
- Introduction to Probability, Introduction to Real Analysis I, Introduction to Complex Analysis, Abstract Algebra, Advanced Abstract Algebra
- Introduction to Computing Tools, Introduction to Object-Oriented Programming, Data Structures I and II, Design and Analysis Computer Algorithms

Drake University Aug. 2022 – May 2024

- Introduction to Physics I and II, Modern Physics, Introduction to Methods in Physics, Error Theory, Descript Astronomy, Astronomical Techniques
- Calculus I-IV, Linear Algebra, Ordinary and Partial Differential Equations, Math Reasoning (Discrete Math), Numerical Analysis

Outreach, Leadership, and Teaching Experience

Mentorship Opportunities for Research Engagement Volunteer | *Northwestern University* Sep. 2026 - Present
LUC Physics Club President | *Loyola University Chicago* Aug. 2025 – May 2026
LUC Mathematics Supplemental Instructor | *Loyola University Chicago* Jan. 2026 – May 2026
LUC Physics Department Homework Grading | *Loyola University Chicago* Aug. 2025 – May 2026
Physics Department Office Assistant | *Loyola University Chicago* Aug. 2025 – May 2026
Physics Department Mentorship Program | *Loyola University Chicago* Aug. 2025 – May 2026
LUC Mathematics Peer Tutor | *Loyola University Chicago* Aug. 2025 – Dec. 2025
Physics Club Tutor | *Loyola University Chicago* Aug. 2024 – May 2026
Drake University Mathematics Tutor | *Drake University* Jan. 2024 – May 2024
Drake Municipal Observatory Volunteer | *Drake University* Aug. 2022 – May 2024

Technical Skills

Languages: Python, C, C++, C#, Java, Fortran, LaTeX, Unix, Red Hat Linux, PyTorch, Scikit-Learn
Developer Tools: Git, VS Code, PyCharm, IntelliJ, SLURM
Engineering Tools/Skills: Fusion360, UltimakerCura, PrusaSlicer, LTSpice, Soldering

Extracurricular Activities

Keynotes Graduate a cappella | *Northwestern University* Sep. 2026 – Present
Loyolacappella (Treasurer) | *Loyola University Chicago* Aug. 2024 – May 2026
Loyola University Choral | *Loyola University Chicago* Aug. 2024 – May 2026
Brocal Chords | *Drake University* Aug. 2023 – May 2024
Drake Choir & Community Choir | *Drake University* Aug. 2022 – May 2024